

Amanah Care

The care record only your family can read.

Abdul Quayum Adekunle · safio

amanah = a trust you are responsible for

THE DIAGNOSIS

Ammi is 78. Three of her children share her care. Here is the system.

Fatima · 07:52

did she take the morning ones?

Sister A · 08:10

yes. she ate half. who has her this afternoon?

Abdullah · 08:41

I thought Fatima. I have the pharmacy at 11 I think?

Fatima · 09:15

no that was last week. what did Dr Rahman say wednesday??

Sister A · 09:20

scroll up 🙄

- No plan, so things get missed or done twice.
- No handoff, so the next person does not know what happened.
- No record of who did what, so one sibling carries it all.
- And the whole history of her health sits with a chat company.

CARE IS ALREADY HAPPENING. IT IS JUST NOT TRACKED.

1 in 4

American Muslims care for an older adult, versus 16% of the general public.

ISPU, Support and Sacrifice, 2026 (American Muslim Poll 2025)

52%

of Muslims aged 30 to 49 care for an elder. 21% of 18 to 29, 31% of 50 and over.

ISPU, Sandwiched

42%

of Canadians 15 and over gave unpaid care in 2022: 13.4 million people, 1.8 million of them sandwiched.

Statistics Canada, Canadian Social Survey 2022

31% depression, 49% burden among family caregivers of people with dementia, across 43 studies.

Collins & Kishita, Ageing & Society, 2020, meta-analysis

18% vs 7% of Muslim caregivers report difficulty finding culturally appropriate care.

ISPU 2026, cited in the MuslimHacks 2026 Challenge Guide

"Medication, appointments, meals, transport and personal care may be tracked across WhatsApp, memory and separate calendars." Challenge Guide. Health information needs consent and privacy protection under Quebec Law 25, Ontario PHIPA and PIPEDA. A chat group meets none of that.

WhatsApp is a conversation. Care needs a record.

A record has five parts. Existing tools each do one. None do all five with the elder's preferences built in, and none keep the family in control of the data.

Plan

Her routine: meds, meals, prayers, walk, pickups.

Actuals

What was really done, by whom, when.

Handoff

What happened.
What is next.

Load

Who carried the week.

Privacy

Only the family can read it.

Amanah Care: a private care record for families.

The next caregiver always knows what happened and what is next. The family sees who did what. Nobody else can read a word of it.

Routine	Ammi's plan, once. Every day becomes a checklist measured against it.
Log	Two taps: category, what happened. A note if you want.
Hand off	Summary written for you from the log. "What is next" written for you from the routine.
Dashboard	Who has Ammi right now. Meds 2 of 3. Mood, last logged. Next pickup and who is on it. Who carried the week.
Family key	Made on the phone, shared by QR in person, once. After that you log in with your email and password. The server stores scrambled text and a password-locked copy of the key it cannot open.
Ammi's screen	The elder gets her own view, in big type: who is with her, what is coming, her prayers, who can read her record.

One afternoon with Ammi's family, on two phones.

1. **Sister A** opens Home: Ammi's day, 8 of 12 done, meds 2 of 3, next pickup Sunday 11:00, Abdullah.
2. She taps **Done** on dinner. The plan and the tiles update.
3. She opens **Hand off** to Fatima. What happened and what is next are already written. She sends.
4. **Fatima's phone** lights up. She reads the card with Ammi's preferences on top, and accepts.
5. Home now says **Fatima has Ammi**. Ammi's own screen says it in big type. The week's load bar moves.
6. We open the **database**. Every row is scrambled text.

What the judges should watch for

- Nothing typed except one line, everything else composed from the record.
- Two real browsers, one real backend, live handoff in seconds.
- The same data, readable on the phones, unreadable on the server.

WHY THIS BEATS THE FAMILY GROUP CHAT

QUESTION A CAREGIVER ASKS	WHATSAPP GROUP	AMANAHA CARE
What is Ammi's plan today?	In someone's head	Today's plan, 8 of 12 done
Did she take her afternoon meds?	Scroll and ask	Meds 2 of 3, ✓ Sister A 14:05
What do I need to know for tonight?	Whatever got typed	Handoff card: what happened, what is next
Who is taking her to the pharmacy?	"I thought you were"	Sunday 11:00, Abdullah, on the dashboard
Is one of us doing everything?	Invisible until someone burns out	Who carried the week, per person
Her language, diet, prayer, modesty?	Retold every time	On top of every handoff card
Who can read her health history?	Meta, and anyone forwarded	Only phones holding the family key

HOW IT WORKS

Every action is an event. Every event is sealed before it leaves the phone.

1

Phone

Plaintext + family key
H. Encrypts with AES-
GCM, fresh IV each
time.

READABLE

2

API

Node. Checks
membership,
appends. Holds no
key.

SEALED

3

Redis Streams

Append-only event
log. Replayable.

SEALED

4

Projector

Consumer group.
Idempotent by event
id.

SEALED

5

Postgres

4 tables + 1 view.
Read models from
metadata only.

SEALED

6

Family phones

Poll, decrypt, compute
the dashboard locally.

READABLE

Why event sourcing

The dashboard, the routine, the record and the workload were added without touching the write path. They are projections over the same log. Replay rebuilds everything.

Runs in one command

Docker Compose: six containers, web, api, projector, notifier, redis, postgres; a notifier consumer group turns subscribed events into alerts. Fallback: one node process runs the same api and projector on an in-memory Postgres.

Even we, the operators, cannot read Ammi's day.

- The family key is made in the browser. It is never sent anywhere.
- The server keeps a SHA-256 check of the key: enough to prove you hold it, never to read with it.
- Every payload is AES-GCM with a fresh 96-bit IV. Tampering fails to decrypt.
- Members join by QR, face to face, once. Later logins fetch a copy of the key locked with the password. The server cannot open it.
- Clear on the server: event type, category, who, when. Sealed: names, notes, summaries, preferences, routine.
- Workload is counted from metadata. No decryption needed.
- No LLM anywhere in the loop. The server could not run one on this data if it wanted to.
- A test asserts a logged note appears in no server-side row, no stream entry, no API response.

TESTED

40 automated tests. Real routes, real schema, in process.

100%

line coverage

api/app.js · projector/apply.js ·
web/crypto.js

90%

branch coverage

Node's built-in test runner and
coverage. No mocks of our own code.

100%

function coverage

Every exported function exercised.

What the suite checks

- Guards: non-members rejected, only the elder may change consent.
- Stream fields, idempotent replay, handoff lifecycle open → acknowledged.
- AES-GCM round trip, fresh IV per message, wrong key and tampering give null.

How it runs

The real `init.sql` loads into an in-memory Postgres. Requests hit the real Express app. A fake stream records every XADD and pushes it through the real projector before the read models are checked. `npm test`, 2.5 seconds, no Docker.

WHAT IS REAL, WHAT IS SIMPLIFIED, WHAT IS NOT BUILT

REAL

- Create family, join by QR or code, then log in with email or login + password
- Elder view in big type
- Log, routine, today's plan
- Handoff: compose, send, accept
- Dashboard, record, workload
- End-to-end encryption, ciphertext-only server
- Redis Streams → projector → Postgres
- Test suite with coverage
- Alerts: per-member event subscriptions, notifier on the stream
- Login and logout

SIMPLIFIED

- Notifications by polling every 4 s, not push
- Login exists, but the demo default password is 333, with no rate limiting or recovery yet
- One Redis stream for all families, one consumer group
- Support workers hold the same key as family

NOT BUILT

- Elder consent filter (needs a scoped support key)
- Key rotation after a lost phone
- Reminders on open handoffs
- Urdu and Arabic for the elder view
- Native mobile apps

WHO PAYS, AND WHAT IT COSTS TO RUN

Families: free

The record is theirs. Free keeps the WhatsApp-replacement honest and adoption in-person, family to family, masjid to masjid.

Agencies and community care: paid

Home-care agencies and community programs that place support workers need handoff logs and scoped access. Per elder, per month. The support role already exists in the model.

Cost of running: small

Stateless API, a stream, a database. The server holds ciphertext only, so it is not a health-data honeypot. One small VM serves thousands of families.

How we know it works

Plan adherence per day. Handoffs accepted within 30 minutes. Number of active caregivers per family, because the point is spreading the load. Weekly return rate.

WHAT IS NEXT

Monday

- Five Montreal families, Urdu and Arabic speaking, one month pilot
- Urdu and Arabic for the elder's screen, on the tablet on the kitchen table

Next month

- Scoped key for support workers, so the consent filter is cryptographic, not cosmetic
- Push notifications, key rotation, password policy and recovery

Later

- Native apps, offline first
- Voice to structured log, on the phone, opt in per family. AI at the edge, never on the server
- Per-family streams, multi-region

On the phone, where the plaintext already is. Never on the server.

Say it, do not type it

"Ammi ate half her lunch, took her meds, seemed tired" becomes three log items: meal, meds, mood. The log already takes structured items, so a parser plugs in front of it.

Read it her way

Handoff summary in plain sentences, and in Urdu or Arabic for the elder's large-print view. Translation on the device.

Patterns, not predictions

"Refused lunch 3 of the last 5 days." Counted from the record, shown as a fact. The family decides what it means. No diagnosis, no triage.

How, without breaking the privacy claim

Chrome's built-in on-device APIs (Prompt, Summarizer, Translator on Gemini Nano, stable on desktop since Chrome 148) and Apple's on-device models in the native app. Opt in per family. Nothing leaves the phone.

Why not tonight

A cloud model call from the demo would contradict "the server cannot read it" on stage. So it is roadmap, designed to fit the architecture rather than bolted on.

REMEMBER THIS

Care was already happening.
Now it has a record.
And only the family can read it.

We ask Introductions to community care programs and five families for the pilot.

Team Abdul Quayum Adekunle · safio

APPENDIX · TECHNICAL IMPLEMENTATION

Delivery format?	Web app, phone first, any browser. Two windows on stage stand in for two phones. Native mobile is roadmap.
Technologies?	Node/Express API, Redis Streams event log, Postgres read models, vanilla JS + WebCrypto AES-GCM, nginx, Docker Compose. Node's built-in test runner. Vended QR library. No frameworks, no CDNs, so the demo needs no internet.
Behind the scenes?	Phone encrypts → POST /events → membership check → XADD to the stream → projector consumer group → Postgres events + handoffs + workload view; a second consumer group, the notifier, writes alerts per subscriber → phones poll and decrypt. The dashboard is computed on the phone.
Built from scratch here?	Spec, requirements and two audit rounds were planned ahead. All code was written during the hackathon window. QR generation is a vendored library.
Tradeoffs?	Polling over push, a demo default password over real credential policy, one stream over per-family streams, consent filter deferred. Each one on the honesty slide with the reason.

APPENDIX · IMPACT AND USERS

Target user?	Adult children sharing care of an elderly parent. The elder is the subject and owner of the preferences, not the app user. Support workers join with their own role.
Discovery and adoption?	Family to family, by QR, in person, by design. Community care programs and masjid welfare committees as the channel. Agencies bring their families.
How many benefit?	1 in 4 American Muslims already care for an older adult (ISPU 2026). Muslims are 8.7% of Greater Montreal (Census 2021). Start with Montreal families, then any family with a shared care load.
Impact if fully developed?	Fewer missed meds and appointments, a fairer split of the load, and less of the strain ISPU documents. A family-owned health record the family never had.
Success measure?	Plan adherence, handoff acceptance time, active caregivers per family, weekly return rate.

APPENDIX · DEMO, FUNCTIONALITY, BUSINESS

Fully functional or mocked?	The core loop is real end to end: encrypt, stream, project, store, decrypt. Login is real, with a demo default password. Notifications are polling. The consent filter is not built. Nothing on stage is faked.
Most important feature?	The handoff, composed from the record: what happened from the log, what is next from the routine, the elder's preferences on top.
Hardest part?	Keeping the server useful while it can read nothing: deciding which fields stay clear for routing and counting, and building a whole dashboard from client-side decryption.
Business model?	Free for families. Paid per elder per month by home-care agencies and community programs that need handoff logs and scoped access.
Scale beyond the prototype?	Per-family streams, push, scoped keys, password policy, offline. Ciphertext-only storage keeps compliance and hosting light.
Who pays?	Agencies and community organisations. Families optionally for extras like key escrow between two relatives.
What would it take to make it a real product?	Password policy and key recovery, native apps, scoped support keys, push, offline, accessibility, Urdu and Arabic.

APPENDIX · TEAM, PROCESS, DATA AND PRIVACY

How did you divide the work?	Abdul: architecture, event model, crypto, api and projector, dashboard, tests. safio: in-app invite flow, interface cleanup, the flow view that traces a write hop by hop.
What did you learn?	Cut scope early and say so. One workflow done well beats five half done. Privacy shapes every feature, including which ones you cannot build yet.
Differently next time?	Tests from hour one. Push notifications budgeted from the start.
How did you prioritise?	Requirement register of 46 items, two audit rounds on feasibility in 24 hours, then "cut core": the handoff loop first, everything else labelled roadmap.
What data?	Care events, handoff summaries, preferences, routine, member names, all encrypted. Clear: event type, category, member id, timestamps.
Ethical risks?	Support workers currently hold the same key, so hiding categories is UI only. The password-locked key copy is only as strong as the password, and the demo default is 333. Lost key means lost data until rotation ships. No diagnosis, dosage or triage anywhere, and the metrics are counts, never judgments.
Depends on an LLM?	No. By design the server cannot run one on this data.

APPENDIX · FUTURE AND CLOSING

Next if more time?	Urdu and Arabic for the elder view, scoped support key, push, key rotation, password policy and recovery.
Continue after today?	Yes. Five-family pilot in Montreal next week.
First step after today?	Commit and publish the repo, deploy on one small VM, recruit the pilot families through community care contacts.
Different from existing options?	Task apps organise chores, med apps remind, marketplaces find carers. None connect plan, actuals, handoff and load with the elder's preferences, and none let only the family read the data.
Help needed?	Introductions to community care programs and agencies, pilot families, and a designer for the elder view.
Remember most?	The server cannot read it. Open the database and see for yourself.
Why should this win?	It solves the coordination problem the track asked for, with the elder's preferences built in, privacy that is cryptographic rather than a promise, an honest scope, and a test suite behind it.

APPENDIX · SOURCES

- ISPU, *Support and Sacrifice: Muslims Who Care for Older Adults Face Emotional Strain*, 2026, drawing on American Muslim Poll 2025. ispu.org/reports-and-analysis/support-and-sacrifice/
- ISPU, *Sandwiched: Muslims Are Most Likely to Be Caring for Aging Parents and Have School-Aged Children*. ispu.org/muslim-families/
- Statistics Canada, *More than half of women provide care to children and care-dependent adults in Canada, 2022*, The Daily, 8 Nov 2022. www150.statcan.gc.ca/n1/daily-quotidien/221108/dq221108b-eng.htm
- Statistics Canada, *Study: "Sandwiched" between multiple unpaid caregiving responsibilities*, The Daily, 2 Apr 2024. www150.statcan.gc.ca/n1/daily-quotidien/240402/dq240402d-eng.htm
- Collins, R. N. and Kishita, N., *Prevalence of depression and burden among informal care-givers of people with dementia: a meta-analysis*, Ageing & Society, 2020.
- MuslimHacks 2026 Challenge Guide, Elderly Care (research and fact sheets by Process Plus). References EC1 to EC6 therein, including Law 25, PHIPA, PIPEDA and Attum et al., *Cultural Competence in the Care of Muslim Patients and Their Families*, StatPearls.
- Statistics Canada, Census 2021 religion release: Muslims 4.9% of Canada, 8.7% of Greater Montreal (cited in the Challenge Guide).
- Chrome for Developers, *Built-in AI: Prompt, Summarizer and Translator APIs on Gemini Nano*. developer.chrome.com/docs/ai/built-in